

Algebraic Curves

Chapter 8.2. Riemann's Theorem

PROPOSITION.

(Riemann's Theorem) There exists a nonnegative integer g s.t. $\ell[D] \geq \deg D + 1 - g$ for all divisors D .

PROOF.

Define $s[D] = \deg D + 1 - \ell[D]$. We find a lower bound g to this function. Evidently $s[0] = 0$, so g is nonnegative if it exists. Moreover by PROPOSITION 2, 3 if $D \sim D'$, then $s[D] = s[D']$. By QUESTION 8.8 if $D \leq D'$, then $s[D] \leq s[D']$. Let $x \in \mathbb{K} \setminus \mathbb{k}$ and let $Z = (x)_0$. From PROPOSITION 4 there is a smallest integer τ s.t. $s[rZ] \leq \tau + 1$ for all r , and as $rZ \leq (r+1)Z$ therefore $s[rZ] = \tau + 1$ for large enough r . Specifically therefore denote $g = \tau + 1$. By the previous statements it suffices to show that for any divisor D , there is a linearly-equivalent divisor D' and an integer $r \in \mathbb{N}$ s.t. $D' \leq rZ$. For this write,

$$Z = \sum_{\mathbf{p}} n_{\mathbf{p}} \mathbf{p}, \quad D = \sum_{\mathbf{p}} m_{\mathbf{p}} \mathbf{p}$$

We want to say $D' = D - \text{div}[f]$, and thus we need $m_{\mathbf{p}} - \text{ord}_{\mathbf{p}}[f] \leq rn_{\mathbf{p}}$ for all $\mathbf{p}$. Denote $y = x^{-1}$ and define,

$$T = \{\mathbf{p} \in \mathbb{X} \mid m_{\mathbf{p}} > 0, \text{ord}_{\mathbf{p}}[y] \geq 0\}$$

And define,

$$f = \prod_{\mathbf{p} \in T} (y - y[\mathbf{p}])^{m_{\mathbf{p}}}$$

So that $m_{\mathbf{p}} - \text{ord}_{\mathbf{p}}[f] \leq 0$ if $\text{ord}_{\mathbf{p}}[y] \geq 0$. Lastly if $\text{ord}_{\mathbf{p}} < 0$, then $n_{\mathbf{p}} > 0$, so a large enough r suffices. ■

The smallest such g of Riemann's Theorem is the 'genus' of $\mathbb{X}$ or $\mathbb{C}$ or $\mathbb{K}$. We read off several corollaries of Riemann's Theorem,

1. If $\ell[D_0] = \deg D_0 + 1 - g$ and $D \sim D' \geq D_0$, then $\ell[D] = \deg D + 1 - g$
2. By choosing $x \in \mathbb{K} \setminus \mathbb{k}$, we may compute g by taking r sufficiently large in $g = \deg[r(x)_0] - \ell[r(x)_0] + 1$

3. There is an integer N s.t. for all divisors D of degree greater than N . $\ell[D] = \deg D + 1 - g$

The first two are evident from the theorem. For the third choose exactly that D_0 and let $N = \deg D_0 + g$. Then where $\deg D \geq N$, hence $\deg[D - D_0] + 1 - g > 0$, so by Riemann's Theorem $\ell[D - D_0] > 0$. Thus $D - D_0 + \text{div}[f] \geq 0$ for some $f \in \mathbb{K}$, i.e. $D \sim D + \text{div}[f] \geq D_0$, at which our corollary is immediate.

Given that the genus is computed from the nonsingular model of a curve, birationally equivalent curves share the same genus. Since we have a normalization to only ordinary singularities for such a curve, there is a simple formula for computing the genus.

PROPOSITION.

Let $\mathbb{C}$ be a plane curve with only ordinary multiple points, n its degree, and denote $r_{\mathbf{p}} = m_{\mathbf{p}}[\mathbb{C}]$. Then,

$$g = \frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p} \in \mathbb{C}} \frac{r_{\mathbf{p}}(r_{\mathbf{p}}-1)}{2}$$

PROOF.

By the previous corollary it suffices to find D of sufficiently large degree and compute $\ell[D]$. We then use the Residue Theorem to find all effective divisors linearly-equivalent to it. W.l.o.g. by a projective automorphism let $\{Z = 0\}$ intersect $\mathbb{C} =: \{F = 0\}$ at n distinct points $\mathbf{p}_1, \dots, \mathbf{p}_n$. Define again,

$$E = \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1) \mathbf{q}, \quad r_{\mathbf{q}} = r_{\mathbf{f}[\mathbf{q}]} = m_{\mathbf{f}[\mathbf{q}]}[\mathbb{C}]$$

And,

$$E_m = \sum_{i=1}^n m \mathbf{p}_i - E$$

So E_m is a divisor of degree $mn - \sum_{\mathbf{p} \in \mathbb{C}} r_{\mathbf{p}}(r_{\mathbf{p}} - 1)$. Let $\mathfrak{V}_m$ be the vector space of forms G of degree m that are adjoint to F . Given G is adjoint iff $m_{\mathbf{p}}[G] \geq r_{\mathbf{p}} - 1$ for all $\mathbf{p} \in \mathbb{C}$, using CHAPTER 5.2. THEOREM 1,

$$\dim \mathfrak{V}_m \geq \frac{(m+1)(m+2)}{2} - \sum_{\mathbf{p}} \frac{r_{\mathbf{p}}(r_{\mathbf{p}}-1)}{2}$$

With equality for sufficiently large m . Hence define,

$$\varphi: \mathfrak{V}_m \rightarrow L[E_m], \quad G \mapsto G/Z^m$$

So that φ is a linear map that vanishes iff G is divisible by F . We prove that this map is surjective. For this for any $f \in L[E_m]$ write $f = R/S$ where R, S are forms of the same degree. Hence $\text{div}[Z^m R] \geq \text{div}[S] + E$. Thus by Max Noether's Theorem, there exists an equation $Z^m R = AS + BF$, so $R/S = A/Z^m$ in $\mathbb{k}(F)$. Thus $\varphi[A] = f$ (as $A \in \mathfrak{V}_m$

as $\text{div}[A] = \text{div}[Z^m A] - \text{div}[S] \geq E$). With this result the following sequence of vector spaces is exact,

$$0 \longrightarrow \mathfrak{W}_{m-n} \longrightarrow \mathfrak{V}_m \longrightarrow L[E_m] \longrightarrow 0$$

Where $\mathfrak{W}_{m-n}$ is the space of all forms of degree $m - n$ and the map from it is $H \mapsto F H$. Therefore we are able to compute for sufficiently large m ,

$$\ell[E_m] = \deg E_m + 1 - \left(\frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p}} \frac{r_{\mathbf{p}}(r_{\mathbf{p}}-1)}{2} \right)$$

Thus as $\deg E_m$ is also increasing as m increases, rearranging gives us g and our proposition. ■

We read off several corollaries from this characterization.

1. If $\mathbb{C}$ is a plane curve of degree n , then g is less than the formula but computed on $\mathbb{C}$, what we've previously denoted $g^*[\mathbb{C}]$.
2. If $\sum_{\mathbf{p}} r_{\mathbf{p}}(r_{\mathbf{p}}-1)/2 = (n-1)(n-2)/2$, then $\mathbb{C}$ is rational
3. Where E_m is as previously defined, any $h \in L[E_m]$ may be written $h = H/Z^m$ where H is adjoint and of degree m . In particular $\deg E_{n-3} = 2g-2$ and $\ell[E_{n-3}] \geq g$.

The first corollary follows from the fact that g is eventually arrived at from $g^*[\mathbb{C}]$ through our successive quadratic transformations. The second corollary is a restatement of $g = 0$. The third corollary follows from the exact sequence utilized in the proof. Namely if $m < n$ then $\mathfrak{V} = L[E_m]$.

For example lines and conics are rational. Nonsingular cubics have genus one. Singular cubics are rational. As a nonsingular curve of degree n has genus $(n-1)(n-2)/2$, not all curves are birationally equivalent to a nonsingular plane curve. For instance $Y^2 X Z = X^4 + Y^4$ has one node so is genus two, but no nonsingular curve has genus two.

Question 8.14

Part A

We compute the genus of $F = X^2Y^2 - Z^2(X^2 + Y^2)$. First,

$$\nabla F = (2X(Y^2 - 2Z^2), 2Y(X^2 - 2Z^2), -2Z(X^2 + Y^2))$$

The first coordinate vanishes if $X = 0$ or $Y^2 = 2Z^2$. In the first case $X = 0$ to vanish the second coordinate either $Y = 0$ or $Z = 0$. In the first subcase $X = Y = 0$ of this we are singular at $[0 : 0 : 1]$. In the second subcase $X = Z = 0$ we are singular at $[0 : 1 : 0]$. In the second case $Y^2 = 2Z^2$ either $Y = 0$ to vanish the second coordinate, at which our point is $[0 : 0 : 0] \notin \mathbb{P}^2$, or $X^2 = 2Z^2$, so $X^2 = Y^2 = 2Z^2$. However this choice would not vanish the third coordinate unless all were zero. Thus our singularities are at $[0 : 0 : 1]$ and $[0 : 1 : 0]$. In particular on their affine covers,

$$F[X, Y, 1] = X^2Y^2 - X^2 - Y^2 = -(X - iY)(X + iY) + X^2Y^2$$

$$F[X, 1, Z] = X^2 - X^2Z^2 - Z^2 = (X - Z)(X + Z) - X^2Z^2$$

So the two points are both of multiplicity two, and are simple. Our computation is thus simple,

$$g = \frac{(4-1)(4-2)}{2} - \frac{2(2-1) + 2(2-1)}{2} = 1$$

Part B

We compute the genus of $F = (X^3 + Y^3)Z^2 + X^3Y^2 - X^2Y^3$. We first check the affine cover $Z \neq 0$ for

$$F[X, Y, 1] = X^3 + Y^3 + X^3Y^2 - X^2Y^3 = X^3(1 + Y^2) + Y^3(1 - X^2)$$

The lowest form is squarefree cubic so the singularity at $\mathbf{0}$ is ordinary and order three. There are no other singularities on this cover. Now on the affine cover $Y \neq 0$,

$$F[X, 1, Z] = (X^3 + 1)Z^2 + X^3 - X^2 = Z^2 - X^2 + X^3 + X^3Z^2$$

The lowest form is squarefree quadratic so the singularity at $\mathbf{0}$ is ordinary and order two. There are no other singularities on this cover. Lastly on the affine cover $X \neq 0$,

$$F[1, Y, Z] = (1 + Y^3)Z^2 + Y^2 - Y^3 = Z^2 + Y^2 - Y^3 + Y^3Z^2$$

The lowest form is squarefree quadratic so the singularity at $\mathbf{0}$ is ordinary and order two. There are no other singularities on this cover. Given all singularities are ordinary we may use the formula to compute,

$$g = \frac{(5-1)(5-2)}{2} - \frac{3(3-1) + 2(2-1) + 2(2-1)}{2} = 1$$

Question 8.15

PROPOSITION.

Let $D = \sum_{\mathbf{p}} n_{\mathbf{p}} \mathbf{p}$ be an effective divisor. Define,

$$S = \{\mathbf{p} \in \mathbb{X} \mid n_{\mathbf{p}} > 0\}$$

And $\mathbb{U} = \mathbb{X} \setminus S$. Then $L[rD] \subseteq \mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ for all $r \in \mathbb{N}$.

PROOF.

By the definition of $\mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ as rational functions on $\mathbb{X}$ which does not have poles on $\mathbb{U}$, it suffices to prove that each element of $L[rD]$ does not have poles on $\mathbb{U}$. Assume for contradiction there were $z \in L[rD]$ s.t. it had a pole on $\mathbb{U}$ at some $\mathbf{q}$, i.e. $\text{ord}_{\mathbf{q}}[z] < 0$. Being in this space means that $0 \leq rD + \text{div}[z]$. However at $\mathbf{q}$, given $n_{\mathbf{q}} = 0$ as $\mathbf{q} \in \mathbb{U} = S^C$ and D is effective,

$$0 \leq 0 + \text{ord}_{\mathbf{q}}[z] < 0$$

Which is a contradiction. Thus elements of $L[rD]$ cannot have poles on $\mathbb{U}$, which directly implies our proposition. ■

Question 8.16

PROPOSITION.

Let $\mathbb{U}$ be an arbitrary open nonempty proper subset of $\mathbb{X}$. Then $\mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ is infinite-dimensional.

PROOF.

Let g be the genus of $\mathbb{X}$, and choose g points $\mathbf{x}_i$ (perhaps with multiplicity) on it. Define,

$$D = \sum_i 1\mathbf{x}_i$$

By QUESTION 8.15 $L[rD]$ is a subset of $\mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ for all $r \in \mathbb{N}$. To see that this space is not trivial, by Riemann's Theorem we have the bound $\ell[rD] \geq rg + 1 - g \geq 1$ so there is at least one rational function in it. Thus assume for contradiction there exists a basis $e_1, \dots, e_n$ of $\mathcal{O}_{\mathbb{U}}[\mathbb{X}]$, so that,

$$\mathcal{O}_{\mathbb{U}}[\mathbb{X}] = \mathbb{k}e_1 + \dots + \mathbb{k}e_n$$

Let $\mathbf{p}$ be a pole of an element $z \in L[D]$. Observe that the divisor of an arbitrary element of $\mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ will therefore have order here,

$$\text{ord}_{\mathbf{p}} \mathcal{O}_{\mathbb{U}}[\mathbb{X}] = \text{ord}_{\mathbf{p}} [\mathbb{k}e_1 + \dots + \mathbb{k}e_n] \geq \min_{1 \leq i \leq n} \text{ord}_{\mathbf{p}} [e_i] =: N$$

However by taking r sufficiently large s.t. $r \text{ord}_{\mathbf{p}} [z] < N$ we are given an element $z^r \in L[rD] \subseteq \mathcal{O}_{\mathbb{U}}[\mathbb{X}]$ that cannot be built from this basis, a contradiction.

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Question 8.17

PROPOSITION.

Let $\mathbb{X}, \mathbb{Y}$ be nonsingular projective curves and $f: \mathbb{X} \rightarrow \mathbb{Y}$ a dominant morphism. Then $f[\mathbb{X}] = \mathbb{Y}$.

PROOF.

Assume for contradiction there exists $\mathbf{p} \in \mathbb{Y} \setminus f[\mathbb{X}]$, i.e. $f[\mathbb{X}] \subset \mathbb{Y} \setminus \{\mathbf{p}\}$. By pullback we have the relation $f^* \mathbb{k}[\mathbb{Y} \setminus \{\mathbf{p}\}] \subset \mathbb{k}[\mathbb{X}] = \mathbb{k}$. Now $\mathbb{Y} \setminus \{\mathbf{p}\}$ is an open subset of $\mathbb{Y}$, so by QUESTION 8.16 its coordinate ring is an infinite-dimensional vector space over $\mathbb{k}$. Thus the pullback $f^* \mathbb{k}[\mathbb{Y} \setminus \{\mathbf{p}\}]$ is an infinite-dimensional vector space over $\mathbb{k}$ too (by taking the injective by dominant morphism pullback of the basis). However our inequality presents that it is bounded above by a one-dimensional vector space, a contradiction.

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Question 8.18

PROPOSITION.

A morphism from a projective curve $\mathbb{X}$ to a curve $\mathbb{Y}$ is either constant or surjective. If it is surjective, then $\mathbb{Y}$ must be projective.

PROOF.

Assume there exists $\mathbf{p} \in \mathbb{Y} \setminus f[\mathbb{X}]$. Define $\mathbb{U} = \mathbb{Y} \setminus \{\mathbf{p}\}$. By pulling back we have the relation,

$$f^* \mathbb{k}[\mathbb{U}] \subset \mathbb{k}[\mathbb{X}] = \mathbb{k}$$

Moreover w.r.t. the Zariski topology $\mathbb{U}$ is open in $\mathbb{Y}$, so that $\mathcal{O}_{\mathbb{U}}[\mathbb{Y}]$ is infinite-dimensional over $\mathbb{k}$, i.e. spanned by a basis $\{e_i\}_{i=1}^{\infty}$. There are then two cases. If the pullback of this basis $\{e_i \circ f\}_{i=1}^n$ is s.t. $n > 0$ then the aforementioned relation is a contradiction. Hence the morphism must be surjective. If however $n = 0$, then f^* is the trivial homomorphism, so f must be the trivial morphism. Therefore f must indeed be either surjective or constant in the respective cases. In the case that it is surjective then $\mathbb{Y}$ is the image under a morphism of a projective curve so is projective. ■

Question 8.19

PROPOSITION.

Let $f: \mathbb{C} \rightarrow \mathbb{V}$ be a morphism from a projective curve $\mathbb{C}$ to a variety $\mathbb{V}$. Then $f[\mathbb{C}]$ must be a closed subvariety of $\mathbb{V}$.

PROOF.

It suffices to prove $f[\mathbb{C}]$ is equal to its closure $\mathbb{C}'$ in $\mathbb{V}$. Observe that f restricts on here to a dominant rational map $\mathbb{C} \rightarrow \mathbb{C}'$, and so being first the image of a variety we see that $\mathbb{C}'$ is irreducible, and by the morphism $\dim \mathbb{C}' \leq \dim \mathbb{C} = 1$. Hence $\mathbb{C}'$ is either a curve or a point. If it is a point then we are done, only the closure of a point is a point so $f[\mathbb{C}] = \mathbb{C}'$ necessarily. Thus assume $\mathbb{C}'$ is a curve. By the result of QUESTION 8.18 the morphism $f: \mathbb{C} \rightarrow \mathbb{C}'$ of curves must be either constant or surjective. In the case it is constant then $f[\mathbb{C}]$ is automatically a closed subvariety of $\mathbb{V}$. If it is surjective then $f[\mathbb{C}] = \mathbb{C}'$ as desired.

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Question 8.20

PROPOSITION.

Let $\mathbb{C} = \{(X^3 + Y^3)Z^2 + X^3Y^2 - X^2Y^3 = 0\}$ and $\mathbf{p}$ a simple point on it. Let $\mathbb{U} = \mathbb{C} \setminus \{\mathbf{p}\}$. Then there exists nonconstant $z \in \mathbb{k}[\mathbb{U}]$ with $\text{ord}_{\mathbf{p}}[z] \geq -12$.

PROOF.

Note that $\mathbb{C}$ has genus $g = 1$ and is singular, so let $\varphi: \mathbb{X} \rightarrow \mathbb{C}$ its smooth model. In particular it has singularities,

$$m_{[0:0:1]} = 3, \quad m_{[0:1:0]} = 2, \quad m_{[1:0:0]} = 2$$

On $\mathbb{X}$ define, denoting $\mathbf{q} = \varphi^{-1}[\mathbf{p}]$ and $\mathbf{q}_i$ the sites of the singularities,

$$D = 12\mathbf{q} - \sum_i (m_{\mathbf{p}_i} - 1)\mathbf{q}_i$$

Now $\deg D = 12 - 6 - 2 - 2 = 2$. By Riemann's Theorem $\ell[D] \geq 2 + 1 - 1 = 2 > 0$, so this space is nonempty and nontrivial. Namely for z to be in $L[D]$ we require $\text{ord}_{\mathbf{p}}[z] = \text{ord}_{\mathbf{q}}[z] \geq -12$. Now the only pole of any element of $L[D]$ may only be at $\mathbf{q}$, so that $L[D] \subseteq \mathbb{k}[\mathbb{X} \setminus \{\mathbf{q}\}]$. In particular at each site $\mathbf{q}_i$ we also require that $\text{ord}_{\mathbf{q}_i}[z] \geq m_{\mathbf{p}_i} - 1$, so the z of note is also nontrivial. Elsewhere on $\varphi^{-1}[\mathbb{U}]$ it is regular. Thus by CHAPTER 7.5. LEMMA 3.,

$$z \in \bigcap_{\mathbf{x} \in \mathbb{U}} \mathcal{O}_{\mathbf{x}}[\mathbb{C}] = \mathbb{k}[\mathbb{U}]$$

As desired. ■

Question 8.2I

PROPOSITION.

Where $C_0[\mathbb{X}]$ is the divisor class group of $\mathbb{X}$, $C_0[\mathbb{X}] = 0$ iff $\mathbb{X}$ is rational.

PROOF.

($\Rightarrow$)

Recall that $0 = C_0[\mathbb{X}] = D_0[\mathbb{X}]/P[\mathbb{X}]$ implies $D_0[\mathbb{X}] = P[\mathbb{X}]$, i.e. that all divisors of degree zero are rational polynomials of $\mathbb{k}[\mathbb{X}]$. We prove that $g = 0$. Consider the point divisor,

$$D = 1\mathbf{p}$$

For some point $\mathbf{p} \in \mathbb{X}$. By Riemann's Theorem $\ell[D] \geq 1 + 1 - g = 2 - g$. Elements of $L[D]$ are $z \in \mathbb{K}$ that may have at most a simple pole at $\mathbf{p}$. Hence we prove that $\ell[D] = 2$, i.e. that $L[D] = \text{span}\{1, z\}$. Assume for contradiction the contrary, i.e. there exists $z' \in \mathbb{K}$ s.t. z' is not in $\text{span}\{1, z\}$. If z' does not have a simple pole at $\mathbf{p}$ then it must be constant, at which it is 1. Hence assume it has a simple pole. We are now able to write,

$$\text{div}[z] = 1\mathbf{q}' - 1\mathbf{p}, \quad \text{div}[z'] = 1\mathbf{q} - 1\mathbf{p}$$

For some $\mathbf{q}, \mathbf{q}' \in \mathbb{X}$. Consider the divisor $1\mathbf{q} - 1\mathbf{q}'$ of degree zero. By hypothesis there exists a rational polynomial $z'' \in \mathbb{K}$ with this as its divisor. Then $\text{div}[z'] + \text{div}[z''] = \text{div}[z]$, so $z = z' z''$. Thus z has a pole at $\mathbf{q}'$. This contradicts it being in $L[D]$.

($\Leftarrow$)

Let $\mathbb{X}$ be rational. It is therefore isomorphic to $\mathbb{P}^1$. From QUESTION 8.6 $C_0[\mathbb{P}^1] = 0$.

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